Statistical methods for computer science

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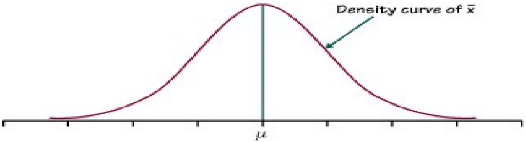
HYPOTHESIS TESTING

Examples

We are making inference about the mean of the distribution of bank deposits of Italian families. Last year, we had $\mu = 3$ (tens of thousands of euro). We want to know if such mean this year is the same or more by knowing that $\sigma = 1$. To this end, we draw a random sample: 5,3,4,3,5,4,4,5,3. The estimated mean is $\bar{y} = 4$.



The difference between 3 and the estimate could be due to:

- sampling error;
- the fact that our idea is wrong.

How could we decide? . . . We use a significance test

Significance test

- A significance test is a method of using data to summarize the evidence about a hypothesis
- In other terms, it is a method to check whether or not the data support certain statements or predictions
- These statements are hypotheses about population.
- A *significance test* about a hypothesis has four elements

1. **Assumptions**
2. **Hypotheses**
3. **Test Statistic**
4. **p-value and conclusion**

Assumptions

Each significance test makes certain assumptions or has certain conditions to be satisfied in order to be valid. These entail

- Type of data $\rightarrow$ Each significance test applies for either quantitative or categorical data
- Randomization $\rightarrow$ Each significance test assumes that the data gathering employed randomization, such as a simple random sample or a randomized experiment.
- Population distribution $\rightarrow$ Some significance tests assume that the characteristic of interest in the population follows a particular distributions, such as the normal, the Bernoulli, etc..
- Sample size $\rightarrow$ Some significance tests, such as those based on the CLT, are valid only when the sample size is sufficiently large

Hypothesis

A hypothesis is a statement about a population, usually of the form that a certain parameter takes a particular numerical value or falls in a certain range of values

Each significance test has two hypotheses

- The null hypothesis (H_0) is a statement that the parameter takes a particular value. It has a single parameter value.
- The alternative hypothesis (H_a or H_1) states that the parameter falls in some alternative range of values. It specifies how the null can be false

In our example we have $H_0 : \mu \leq 3$ vs $H_a : \mu > 3$.

The null and the alternative hypothesis are such that

- They do not overlap $\rightarrow H_0 \cap H_1 = \emptyset$
- Their union is the parameter space (PS) $\rightarrow H_0 \cup H_1 = \mathrm{PS}$

In a significance test, the null hypothesis is presumed to be true unless the data give strong evidence against it

Formally, we have to decide, on the basis of the information contained in the sample, if H_0 has to be rejected or not

Test Statistic

In order to decide if the null should be rejected or not, we summarize the evidence against the null by computing a *(test) statistic*.

A test statistic describes (measures) how far the point estimate falls from the parameter value given in the null hypothesis (usually in terms of the number of standard errors between the two)

- The test statistic is zero when the estimate is equal to the value specified in the null
- If the test statistic falls far from zero in the direction specified by the alternative hypothesis, there is evidence against the null and in favor of the alternative hypothesis

In our example, the test statistic is

$$Z = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}}$$

Z is a random variable such that, if H_0 is true, $Z \sim N(0, 1)$

Based on the data, the observed value of the the test statistic is

$$z_{obs} = \frac{4-3}{1/\sqrt{9}} = 3$$

Is it a large or small value? Should we reject H_0 or not?

p-value

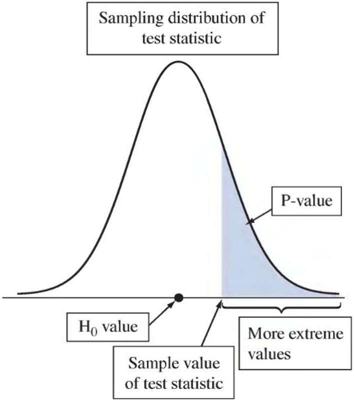
To interpret a test statistic value, we use a probability summary of the evidence against the null hypothesis, H_0

- First, we presume that H_0 is true
- Next, we consider the sampling distribution of the test statistic

We summarize how far out in the tail of this sampling distribution the test statistic falls by computing, when H_0 is true, the probability of that value and values even more extreme. This probability is called p-value

- The smaller the p-value, the stronger the evidence data provide against the null hypothesis
- That is, a small p-value indicates a small likelihood of observing the sampled results if the null hypothesis were true

In our example, p-value=P(Z $>$ 3)=1-P(Z$<$3)=1-0.9987=0.0013.



Sampling distribution of test statistic
P-value
H_0 value
Sample value of test statistic
More extreme values

Conclusion

Report and interpret the p-value in the context of the study.

Based on the p-value, make a decision about H_0 (either reject or do not reject H_0 if the p-value is small or not)

Before seeing the data, we decide how small the p-value would need to be to reject H_0. This cutoff point is called the significance level.

In practice, the most common significance level is 0.05 (or 0.01 if the sample size is very large).

When we reject H_0 we say the test is statistically significant

In our example, we have p-value$=0.0013<0.05$, then we reject H_0

Significance Tests About Proportions

Example: Are Astrologers' Predictions Better Than Guessing?

An astrologer prepares horoscopes for 116 adults.

Each subject also filled out a California Personality Index (CPI) survey.

For a given adult, his or her horoscope is shown to the astrologer along with their CPI survey as well as the CPI surveys for two other randomly selected adults. The astrologer is asked which survey is the correct one for that adult.

In the actual experiment, the astrologers were correct with 40 of their 116 predictions (a success rate of $0.345 > 0.333$).

Assumptions

- The variable is categorical in 0-1 coding
- The sample size is sufficiently large so that the sampling distribution of the sample proportion is approximately normal.

In our example, we have 1 if the astrologer gives a correct answer and 0 otherwise.

Hypotheses

- Initially, the **null hypothesis** has the form $H_0 : p = p_0$.
- The **alternative hypothesis** has the form:
  - $H_1 : p < p_0$ (one-sided, left tail, lower tailed)
  - $H_1 : p > p_0$ (one-sided, right tail, upper tailed)
  - $H_1 : p \neq p_0$ (two-sided, two tails, two tailed)

In our first example we then set $H_0 : p \leq 1/3$ and $H_1 : p > 1/3$ (the predictive power of the astrologer is effective).

Test Statistic

The test statistic is:

$$Z = \frac{\hat{P} - p_0}{\sqrt{p_0(1 - p_0)}}\sqrt{n}$$

It has a standard normal distribution when the null hypothesis is true.

In our example, the observed value of the test statistic is

$$z_{obs} = \frac{0.345 - 0.333}{\sqrt{0.333(1 - 0.333)}}\sqrt{116} = 0.27$$

p-value

The p-value summarizes the evidence against H_0. It describes how unusual the observed data would be if H_0 were true.
There are three ways to compute the p-value:

- $H_1 : p < p_0$ (one-sided, left tail)
$$\Rightarrow \mathrm{P}(Z < z_{obs})$$
- $H_1 : p > p_0$ (one-sided, right tail)
$$\Rightarrow \mathrm{P}(Z > z_{obs})$$
- $H_1 : p \neq p_0$ (two-sided, two tails)
$$\Rightarrow \mathrm{P}(|Z| > |z_{obs}|)$$
$$= \mathrm{P}(Z < -|z_{obs}|) + \mathrm{P}(Z > |z_{obs}|) = 2\mathrm{P}(Z > |z_{obs}|)$$

In our example, p-value$=\mathrm{P}(Z > 0.27)=$1-0.6064=0.3936

Conclusion

Compare the *p*-value with the pre-specified significance level (usually 0.05).

In our example,

- the *p*-value of 0.39 is greater than the significance level
- We do not reject the null hypothesis
- There is *not* strong enough evidence that astrologers have special predictive powers

Example

We want to test if a coin is fair or not. To this end, we toss the coin 121 times obtaining 55 heads. What can we conclude?

- **Assumptions**
  The assumptions are satisfied:
  $121 \times 0.5 = 121 \times (1 - 0.5) = 60.5 > 15$
- **Hypotheses**
  $H_0 : p = 0.5$ vs $H_1 : p \neq 0.5$
- **Test Statistic**
  $$z_{obs} = \frac{55/121 - 0.5}{\sqrt{0.5(1 - 0.5)}}\sqrt{121} = -1$$
- **p-value**
  p-value$= 2 \times P(Z > 1) = 2 \times (1 - 0.8413) = 0.3174$
- **Conclusion**
  We do not reject the null because p-value>0.05

Significance Tests About the Mean

Assumptions

1. The variable is continuous
2. The population distribution is approximately normal
3. The sample size is sufficiently large that the sampling distribution of the sample mean is approximately normal.

Hypotheses

- The **null hypothesis** has the form $H_0 : \mu = \mu_0$
- The **alternative hypothesis** has the form:
  - $H_1 : \mu < \mu_0$ (one-sided, left tail, lower tailed)
  - $H_1 : \mu > \mu_0$ (one-sided, right tail, upper tailed)
  - $H_1 : \mu \neq \mu_0$ (two-sided, two tails, two tailed)

Test Statistic

Measures how far the sample mean falls from the null hypothesis value μ_0, as measured by the number of standard errors between them

A) Known variance

The test statistic is

$$Z = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}}$$

- Under the null hypothesis, it is distributed as a standard Gaussian

B) Unknown variance

The test statistic is

$$T = \frac{\bar{Y} - \mu_0}{S/\sqrt{n}}$$

- Under the null hypothesis, it is distributed as a Student t with $n-1$ degree of freedom (if assumption 2. is true)
- If n is large enough, it has a standard normal distribution even if the population is not approximately normal

p-value

The p-value describes how unusual the test statistics would be if H_0 were true. There are three ways to compute the p-value:

- $H_1 : \mu < \mu_0$ (one-sided, left tail)$\Rightarrow$P$(T < t_{obs})$
- $H_1 : \mu > \mu_0$ (one-sided, right tail)$\Rightarrow$P$(T > t_{obs})$
- $H_1 : \mu \neq \mu_0$ (two-sided, two tails) $\Rightarrow$P$(|T| > |t_{obs}|)$=2P$(T > |t_{obs}|)$

Conclusion

Compare the p-value with a pre-specified significance level (usually 0.05). Reject the null if the p-value is less than the significance level.

Example

In a supermarket, the mean amount of spending per client is 50 Euro. After a restoration, the management wants to test if the new organization induces the clients to increase the amount of spending. In a sample of $n = 196$ of buyers, drawn at random, they recorded $\bar{y} = 52$ and $s = 28$. What can we conclude?

- **Assumptions**
  - The variable (amount of spending) is quantitative.
  - The data are obtained using randomization.
  - The sample size is large enough.
- **Hypotheses**
  $H_0 : \mu \leq 50$ vs $H_1 : \mu > 50$.
- **Test Statistic**
  The test statistic is

$$t_{obs} = \frac{52 - 50}{28/\sqrt{196}} = 1$$

- **p-value**
  p-value=P($Z > 1$)=1-0.8413=0.1587
- **Conclusion**
  We do not reject the null.

Never "Accept H_0" in a Significance Test

- When the observed p-value is larger than the chosen significance level α, we "do not reject H_0"
- It is not appropriate to say "Accept H_0"
- The observed value of the statistic we consider to estimate the parameter of interest (the estimate) is only one of the plausible values it can take
- Similarly, the corresponding p-value is only one out of the possible values it can take (before observing the data, the p-value is a random variable)
- As a result, when it is larger than α, saying "Do not reject H_0" instead of "Accept H_0" emphasizes that the observed p-value is merely one of many plausible values
- The collection of test statistic values for which a significance test rejects H_0 is called the rejection region
- The cutoff point for the rejection region is called the critical value ($\rightarrow c$)

Decisions and types of errors in significance tests

Because of sampling variability, decisions in significance tests always have some uncertainty. A decision can be in error

Two are the possible errors:

	H_0 true	H_0 false
Accept H_0	Correct Decision	Type error II
Reject H_0	Type error I	Correct Decision

For every test, it is possible to compute the probability of a type I or II error.

- The probability of a type I error is denoted with α and this directly corresponds to the significance level we choose for the test

$$\alpha = \Pr(R\, H_0 \mid H_0 \text{ true})$$

- The more serious the consequences of a type I error, the smaller α should be

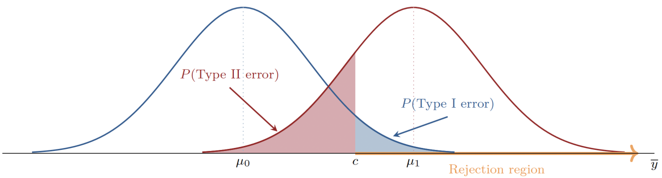
- The probability of a type II error is denoted with β

$$\beta = \Pr(\bar{R}\, H_0 \mid H_1 \text{ true})$$

Optimal test

The ideal test would be that minimizing both α and β

However, the two error probabilities are inversely related. It follows that it is not possible to minimize both simultaneously



- The red curve is the sampling distribution of the estimator when a particular value μ_1 specified in H_1 is the actual value of the mean μ of a Gaussian population
- The rejection region consists of $\bar{y}$ values above some critical value c, depending on the selected α (i.e., the blue-shaded area in the figure)
- The P(Type II error) is the probability to the left of the critical value c under the red curve (i.e., the red-shaded area in the figure)
- When we make α smaller, we move the critical value c to the right, but then P(Type II error) increases

How do we choose the best test?

Two approaches:

1. the best test minimizes a weighted mean of the two error probabilities;
2. given an upper bound for the probability of a type I error, the best test minimizes the probability of a type II error for each possible value of the parameter in the region of the alternative hypothesis (Neyman-Pearson approach)

The tests that we have discussed before follow the second approach

Duality between Significance Tests and Confidence Intervals

A test decision at a particular α-level relates to a confidence interval with the same error probability

The α-level is both P(Type I error) for the test and the probability that the confidence interval method yields an interval that does not contain the parameter

Example

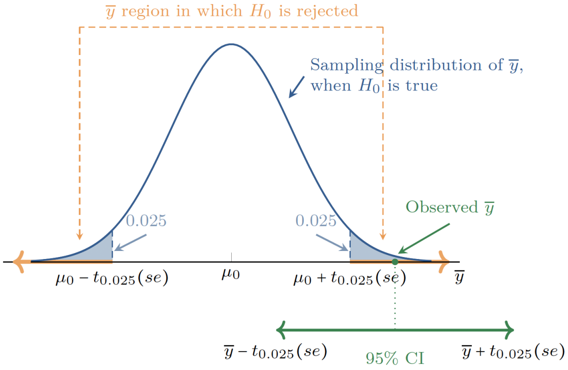
- Consider the significance test

$$H_0 = \mu = \mu_0 \quad vs \quad H_1 : \mu \neq \mu_0$$

  at the significance level $\alpha = 0.05$
- We do reject H_0 when the observed value of the test statistic

$$t = (\bar{y} - \mu_0)/se$$

  is greater, in absolute value, than $t_{0.025,n-1}$
- That is, we reject H_0 when $\bar{y}$ falls more than $t_{0.025}(se)$ far from μ_0
- But if this happens, then the 95% confidence interval for μ, which is $\bar{y} \pm t_{0.025}(se)$ does not contain μ_0, as shown in the subsequent figure


$\overline{y}$ region in which H_0 is rejected
Sampling distribution of $\overline{y}$, when H_0 is true
0.025
0.025
Observed $\overline{y}$
$\mu_0 - t_{0.025}(se)$
μ_0
$\mu_0 + t_{0.025}(se)$
$\overline{y}$
$\overline{y} - t_{0.025}(se)$
95% CI
$\overline{y} + t_{0.025}(se)$